24Cd3014_H_Adithya_CNN_Lab_8

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Note - Sir after monday's class i talked to you about my early submission and you told me to add a comment about the same for the future submission

4 *Evironment Setup & Data Pipeline (Task - 1)*

4.1 *Imorting Librariers*

[1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
import matplotlib.cm as cm
import seaborn as sns
from sklearn.metrics import confusion_matrix, classification_report,f1_score
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers, models,callbacks,regularizers
from tensorflow.keras.datasets import mnist, cifar10
from tensorflow.keras.applications import VGG16
from tensorflow.keras.applications.vgg16 import preprocess_input as␣
↪vgg_preprocess
from scipy.ndimage import rotate as scipy_rotate
from PIL import Image as PILImage
import warnings
warnings.filterwarnings('ignore')
```

4.2 *Checking for gpu*

[2]:
```
# Setting the seeds fo data
np.random.seed(42)
tf.random.set_seed(42)
import random
random.seed(42)
gpu = tf.config.experimental.list_physical_devices('GPU')
if(gpu):
  print(f"Gpu detected : {[g.name for g in gpu]}")
else:
  print("No Gpu deteced")

print(f"Why GPU is better than CPU?")
print(f"Reason\nA cpu will work slower than a gpu because gup can do paralle␣
↪computing whcih helps in fdast compuatation of thousadans of␣
↪matrixmultiplication involved in CNN.")
```

```
Gpu detected : ['/physical_device:GPU:0']
Why GPU is better than CPU?
Reason
A cpu will work slower than a gpu because gup can do paralle computing whcih
helps in fdast compuatation of thousadans of matrixmultiplication involved in
CNN.
```

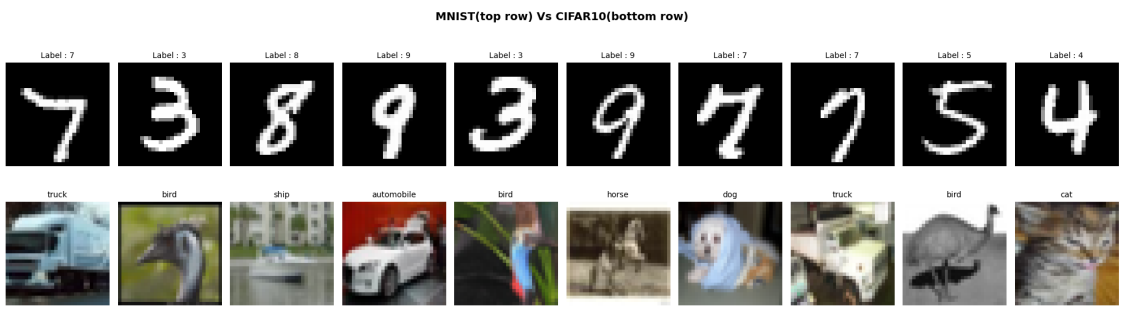
4.3 *Dataset Exploration*

[3]:
```
(mnist_train_x,mnist_train_y),(mnist_test_x,mnist_test_y) =  mnist.load_data()
(cifar10_train_x,cifar10_train_y),(cifar10_test_x,cifar10_test_y) =  cifar10.
↪load_data()

# MNIST data set info
print(f'Info about MNIST data set:')
print(f'image train_shape: {mnist_train_x.shape}')
print(f'label train_shape: {mnist_train_y.shape}')
print(f'image test_shape: {mnist_test_x.shape}')
print(f'label test_shape: {mnist_test_y.shape}')
print (f"Pixel value range: [{int(mnist_train_x.min()),int(mnist_train_x.
↪max())}]")

print("\n")

# CIFAR10 data set info
print(f'Info about CIFAR10 data set:')
print(f'image train_shape: {cifar10_train_x.shape}')
print(f'label train_shape: {cifar10_train_y.shape}')
print(f'image test_shape: {cifar10_test_x.shape}')
```

```
print(f'label test_shape: {cifar10_test_y.shape}')
print (f"Pixel value range: [{int(cifar10_train_x.min()),int(cifar10_train_x.
 ↪max())}]")

# Checking the class balace of MNIST data
print("\nThe class balance of MNIST data set")
uni , cnt = np.unique(mnist_train_y,return_counts=True)
print(f'Class : {uni}')
print(f'Count : {cnt}')
print(f"The class is mostly well balanced except for digits as there are 60k␣
 ↪samples in tarin data with 10 classes and all the classes have roughly 6k␣
 ↪samples except for the class 1 and 5 ")

# Visualization of data
cifar10_class_names = ['airplane', 'automobile', 'bird', 'cat', 'deer', 'dog',␣
 ↪'frog', 'horse', 'ship', 'truck']

fig,axes = plt.subplots(2,10,figsize=(20,6))
fig.suptitle("MNIST(top row) Vs CIFAR10(bottom␣
 ↪row)",fontsize=14,fontweight='bold')

# Random samples for MNIST data
mnist_ind = np.random.choice(len(mnist_train_x),10,replace=False)
for i,id in enumerate(mnist_ind):
  axes[0,i].imshow(mnist_train_x[id],cmap='gray')
  axes[0,i].set_title(f"Label : {mnist_train_y[id]}",fontsize=10)
  axes[0,i].axis("off")

# Random samples for CIFAR10 data
cifar_ind = np.random.choice(len(cifar10_train_x),10,replace=False)
for i,id in enumerate(cifar_ind):
  axes[1,i].imshow(cifar10_train_x[id])
  axes[1,i].set_title(cifar10_class_names[cifar10_train_y[id][0]],fontsize=10)
  axes[1,i].axis("off")

plt.tight_layout()
plt.savefig("Dataset_Sample.png",dpi=300)
plt.show()
```

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-
datasets/mnist.npz
11490434/11490434             0s
0us/step
Downloading data from https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz
170498071/170498071             4s
0us/step
Info about MNIST data set:
```

```
image train_shape: (60000, 28, 28)
label train_shape: (60000,)
image test_shape: (10000, 28, 28)
label test_shape: (10000,)
Pixel value range: [(0, 255)]


Info about CIFAR10 data set:
image train_shape: (50000, 32, 32, 3)
label train_shape: (50000, 1)
image test_shape: (10000, 32, 32, 3)
label test_shape: (10000, 1)
Pixel value range: [(0, 255)]

The class balance of MNIST data set
Class : [0 1 2 3 4 5 6 7 8 9]
Count : [5923 6742 5958 6131 5842 5421 5918 6265 5851 5949]
The class is mostly well balanced except for digits as there are 60k samples in
tarin data with 10 classes and all the classes have roughly 6k samples except
for the class 1 and 5
```



4.4 *Preprocessing Pipeline*

[4]:
```python
def Preprocessing(imgs,labels,data_name="Dataset"):
  print(f"\n[{data_name}] before preprocessing:")
  print(f"Image shape: {imgs.shape},dtype: {imgs.dtype}")
  print(f"Label shape: {labels.shape},dtype: {labels.dtype}")
  print(f"Pixel range: [{int(imgs.min())},{int(imgs.max())}]")
  print(f"Sample labels: {labels[:5].flatten()}")

  imgs = imgs.astype(np.float32)/255.0

  if imgs.ndim==3:
    imgs = np.expand_dims(imgs,axis=-1)

  labels = labels.flatten()
```

```
  num_classes = 10
  one_hot = np.zeros((len(labels),num_classes),dtype=np.float32)
  one_hot[np.arange(len(labels)),labels] = 1.0

  print(f"\n[{data_name}] after preprocessing:")
  print(f"Image shape: {imgs.shape},dtype: {imgs.dtype}")
  print(f"Label shape: {one_hot.shape}")
  print(f"Pixel range: [{int(imgs.min()):.4f},{int(imgs.max()):.4f}]")
  print(f"Sample one hot labels: {one_hot[0]}")

  return imgs,one_hot

# Preprocessing of MNIST
mnist_tr_x_p,mnist_tr_y_p= Preprocessing(mnist_train_x,mnist_train_y,"MNIST␣
 ↪train")
mnist_te_x_p,mnist_te_y_p= Preprocessing(mnist_test_x,mnist_test_y,"MNIST test")

# Preprocessing of CIFAR10
cifar10_tr_x_p,cifar10_tr_y_p=␣
 ↪Preprocessing(cifar10_train_x,cifar10_train_y,"CIFAR10 train")
cifar10_te_x_p,cifar10_te_y_p=␣
 ↪Preprocessing(cifar10_test_x,cifar10_test_y,"CIFAR10 test")
```

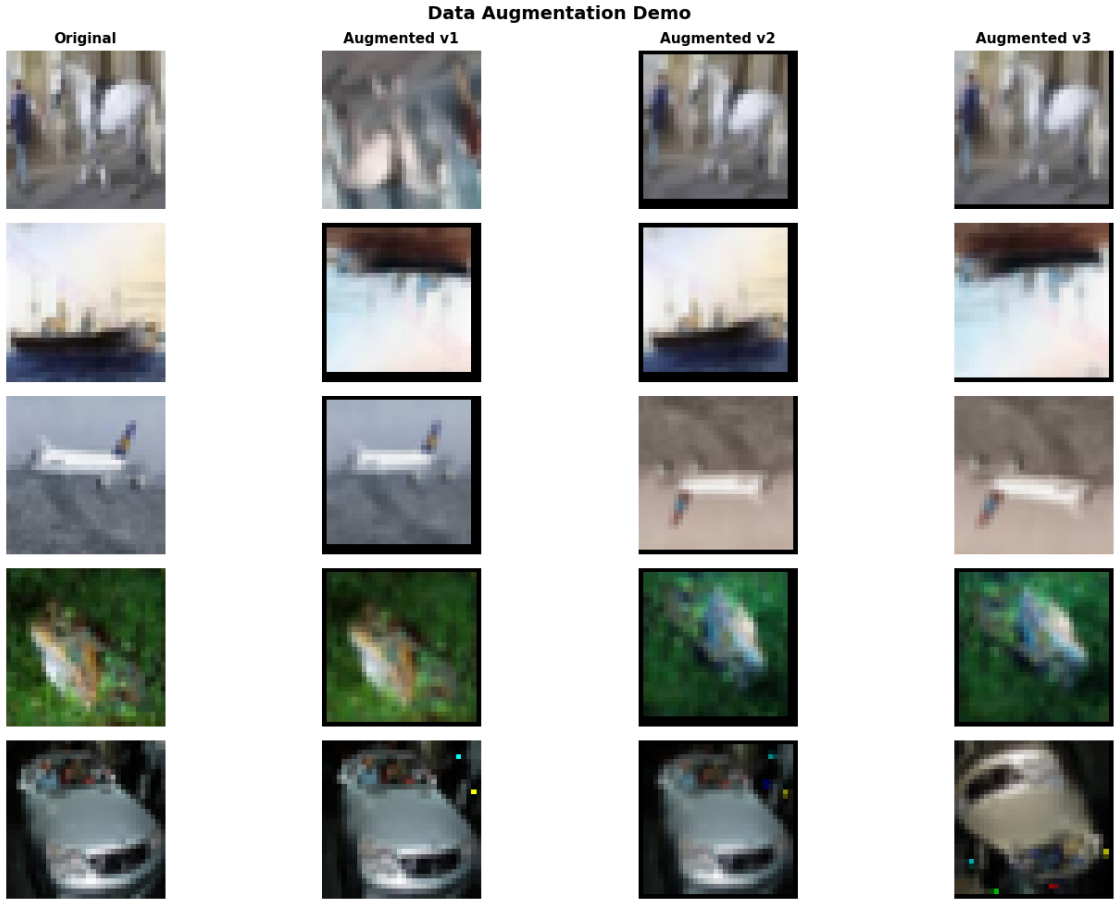
```
[MNIST train] before preprocessing:
Image shape: (60000, 28, 28),dtype: uint8
Label shape: (60000,),dtype: uint8
Pixel range: [0,255]
Sample labels: [5 0 4 1 9]

[MNIST train] after preprocessing:
Image shape: (60000, 28, 28, 1),dtype: float32
Label shape: (60000, 10)
Pixel range: [0.0000,1.0000]
Sample one hot labels: [0. 0. 0. 0. 0. 1. 0. 0. 0. 0.]

[MNIST test] before preprocessing:
Image shape: (10000, 28, 28),dtype: uint8
Label shape: (10000,),dtype: uint8
Pixel range: [0,255]
Sample labels: [7 2 1 0 4]

[MNIST test] after preprocessing:
Image shape: (10000, 28, 28, 1),dtype: float32
Label shape: (10000, 10)
Pixel range: [0.0000,1.0000]
Sample one hot labels: [0. 0. 0. 0. 0. 0. 0. 1. 0. 0.]
```

```
[CIFAR10 train] before preprocessing:
Image shape: (50000, 32, 32, 3),dtype: uint8
Label shape: (50000, 1),dtype: uint8
Pixel range: [0,255]
Sample labels: [6 9 9 4 1]

[CIFAR10 train] after preprocessing:
Image shape: (50000, 32, 32, 3),dtype: float32
Label shape: (50000, 10)
Pixel range: [0.0000,1.0000]
Sample one hot labels: [0. 0. 0. 0. 0. 0. 1. 0. 0. 0.]

[CIFAR10 test] before preprocessing:
Image shape: (10000, 32, 32, 3),dtype: uint8
Label shape: (10000, 1),dtype: uint8
Pixel range: [0,255]
Sample labels: [3 8 8 0 6]

[CIFAR10 test] after preprocessing:
Image shape: (10000, 32, 32, 3),dtype: float32
Label shape: (10000, 10)
Pixel range: [0.0000,1.0000]
Sample one hot labels: [0. 0. 0. 1. 0. 0. 0. 0. 0. 0.]
```

4.5 *Data Augmentation Pipeline*

[5]:
```
def aug_img(image):
  img = image.copy()
  if np.random.rand()<0.5:
    img = np.flip(img)

  angle = np.random.uniform(-10,10)
  img = scipy_rotate(img,angle,reshape=False,mode="nearest")

  zoom_fac = np.random.uniform(0.9,1.1)
  h,w = img.shape[:2]
  new_h = int(h*zoom_fac)
  new_w = int(w*zoom_fac)

  pil_img = PILImage.fromarray((img*255).astype(np.uint8))
  pil_img = pil_img.resize((new_w,new_h),PILImage.BILINEAR)
  img_resized = np.array(pil_img).astype(np.float32)/255.0

  if zoom_fac>1.0:
    st_h = (new_h-h)//2
    st_w = (new_w-w)//2
```

```
    img = img_resized[st_h:st_h+h,st_w:st_w+w]
  else:
    pad_h = (h-new_h)//2
    pad_w = (w-new_w)//2
    img = np.zeros_like(image)
    img[pad_h:pad_h+new_h,pad_w:pad_w+new_w] = img_resized

  return np.clip(img,0.0,1.0)

np.random.seed(42)
sample_ind = np.random.choice(len(cifar10_tr_x_p),5,replace=False)

fig,axes = plt.subplots(5,4,figsize=(15,10))
fig.suptitle("Data Augmentation Demo",fontsize=14,fontweight="bold")
col_labels = ["Original","Augmented v1","Augmented v2","Augmented v3"]

for col,label in enumerate(col_labels):
  axes[0,col].set_title(label,fontsize=11,fontweight="bold")

for r,id in enumerate(sample_ind):
  original = cifar10_tr_x_p[id]
  class_name = cifar10_class_names[cifar10_train_y[id][0]]
  axes[r,0].imshow(original)
  axes[r,0].set_ylabel(class_name,fontsize=10,rotation=0,labelpad=50)
  axes[r,0].axis("off")

  for c in range(1,4):
    aug = aug_img(original)
    axes[r,c].imshow(aug)
    axes[r,c].axis("off")

plt.tight_layout()
plt.savefig("augmentation_demo.png",dpi=300)
plt.show()
```



4.6 Analysis Questions

Q1. What does the channel dimension represent in a tensor of shape (N, H, W, C)? Explain with reference to both a greyscale image and an RGB image.

Ans– The channel dimension C represents the number of independent signal types at each spatial location. For a grayscal image (C=1) Each pixel has a single value representing brightness (0=black, 255=white). Shape: (N, 28, 28, 1)

For an RGB image (C=3) Each pixel has 3 values: how much Red, Green, and Blue light it emits. Shape: (N, 32, 32, 3) Example pixel: [255, 0, 0] = pure red.

In Conv2D, the kernel slides over (H, W) but sees ALL C channels at once, computing a dot product across all three (or one) channel values together.

Q2. CIFAR-10 images are 32×32 pixels. If you were training a CNN on 1024×1024 satellite images, what data loading strategies would you use to avoid running out of memory? Name at least two techniques and explain how each one helps.

Ans– 1. tf.data pipeline with on-the-fly loading: Instead of loading all images into RAM, read from disk one batch at a time. tf.data.Dataset.from_generator() + .prefetch() loads the next batch

while the GPU is training on the current one → no memory bottleneck.

2. Image tiling / patch-based training: Divide each 1024×1024 image into 32×32 or 64×64 tiles. Train on tiles. This reduces each input to CNN-manageable sizes without losing detail. At inference, reassemble predictions from tiles using sliding window.

Q3. Suppose a student applies normalisation to the test set using the mean and standard deviation computed from the test set itself. What is wrong with this approach?

Ans– In production, when a new image arrives, you don't know its statistics you just have the image. The model was trained expecting inputs normalised by the training set's mean/std. If test normalisation uses different values, the distribution shifts and the model gets unexpected inputs.

Correct approach is to compute mean and std of training set only, then apply the same values to both validation and test sets.

5 *Building a CNN from Scratch (Task - 2)*

5.1 *Manual 2D Convolution(Numpy only)*

[6]:
```
def conv2d(img,kernel,stride=1,padding=0):
  h,w = img.shape
  kh,kw = kernel.shape

  if padding>0:
    img = np.pad(img,padding,mode="constant",constant_values=0)
    h_pad,w_pad=img.shape
  else:
    h_pad,w_pad=h,w

  out_h = (h_pad-kh)//stride+1
  out_w = (w_pad-kw)//stride+1

  feature_map = np.zeros((out_h,out_w),dtype=np.float64)

  for i in range(out_h):
    for j in range(out_w):
      row_st = i*stride
      col_st = j*stride
      patch = img[row_st:row_st+kh,col_st:col_st+kw]
      feature_map[i,j] = np.sum(patch*kernel)

  return feature_map



# Testing given a 5X5 img with sobel-X kernel
test_img = np.array([
    [3,1,0,2,4],
```

```
    [1,5,3,2,1],
    [0,2,6,4,3],
    [2,3,1,5,2],
    [1,0,2,3,4]
],dtype=np.float64)
sobel_x = np.array([
    [-1,0,1],
    [-2,0,2],
    [-1,0,1]
],dtype=np.float64)

res = conv2d(test_img,sobel_x,stride=1,padding=0)

print(f"Input image:")
print(test_img.astype(int))
print(f"Sobel-X kernel:")
print(sobel_x.astype(int))
print(f"Convolution Output Iamge:")
print(res)
print(f"\nOutput shape:{res.shape}")
print(f"Expected shape using formula: ({(5-3)//1 + 1},{(5-3)//1 + 1})")
print(f"Interpretation:\nPositive values- left to right bright inc\nNegtaive␣
↪values- right to left brightness dec")
```

```
Input image:
[[3 1 0 2 4]
 [1 5 3 2 1]
 [0 2 6 4 3]
 [2 3 1 5 2]
 [1 0 2 3 4]]
Sobel-X kernel:
[[-1  0  1]
 [-2  0  2]
 [-1  0  1]]
Convolution Output Iamge:
[[ 7. -3. -3.]
 [13.  3. -7.]
 [ 5.  9.  1.]]

Output shape:(3, 3)
Expected shape using formula: (3,3)
Interpretation:
Positive values- left to right bright inc
Negtaive values- right to left brightness dec
```

5.2 *Output Size Derivation*

[7]:
```
def calc_output_size(input_size,kernel,padding,stride):
  output = (input_size-kernel+2*padding)/stride+1
  return int(output)

print("""
formula: Output = floor((Input-kernel + 2 x padding)/stride)+1

a) input=28, kernel=5, padding=0,stride=1
          = floor((28-5+2x0)/1)+1
          = floor(23/1)+1
          = 24
   output = 24x24""")
print(f"   verified:{calc_output_size(28,5,0,1)}x{calc_output_size(28,5,0,1)}")
print("""
b) input=28, kernel=3, padding=0,stride=1
          = floor((28-3+2x1)/1)+1
          = floor(27/1)+1
          = 28
   output = 28x28 same after padding
   Therefore padding keeps spatial size identical.""")
print(f"   verified:{calc_output_size(28,3,1,1)}x{calc_output_size(28,3,1,1)}")

print("""
c) input=32, kernel=3, padding=0,stride=2
          = floor((32-3+2x0)/2)+1
          = floor(29/2)+1
          = 15
   output = 15x15
   stride=2 halves the spatial size.""")
print(f"   verified:{calc_output_size(32,3,0,2)}x{calc_output_size(32,3,0,2)}")

step1=calc_output_size(32,3,1,1)
step2=calc_output_size(step1,3,0,1)
print("""
d) Two consecutive layers on 32x32:
   Layer 1: Input=32,K=3,P=1,S=1
   = floor((32-3+2x1)/1)+1 = 32
   Output = 32x32 (same padding preserves size)
   Layer 2: Input=32,K=3,P=0,S=1
   = floor((32-3+2x0)/1)+1 = 30
   Final Output = 30x30""")
print(f"   verified:{step1}X{step1} -> {step2}X{step2}")
```

```
formula: Output = floor((Input-kernel + 2 x padding)/stride)+1
```

```
a) input=28, kernel=5, padding=0,stride=1
          = floor((28-5+2x0)/1)+1
          = floor(23/1)+1
          = 24
   output = 24x24
   verified:24x24

b) input=28, kernel=3, padding=0,stride=1
          = floor((28-3+2x1)/1)+1
          = floor(27/1)+1
          = 28
   output = 28x28 same after padding
   Therefore padding keeps spatial size identical.
   verified:28x28

c) input=32, kernel=3, padding=0,stride=2
          = floor((32-3+2x0)/2)+1
          = floor(29/2)+1
          = 15
   output = 15x15
   stride=2 halves the spatial size.
   verified:15x15

d) Two consecutive layers on 32x32:
   Layer 1: Input=32,K=3,P=1,S=1
   = floor((32-3+2x1)/1)+1 = 32
   Output = 32x32 (same padding preserves size)
   Layer 2: Input=32,K=3,P=0,S=1
   = floor((32-3+2x0)/1)+1 = 30
   Final Output = 30x30
   verified:32X32 -> 30X30
```

5.3 *Impement LeNet-5*

[8]:
```python
def build_lenet5(input_shape=(28,28,1),num_classes=10):
  model = models.Sequential([
      layers.Input(shape=input_shape),
      layers.Conv2D(6,(5,5),padding="valid",name="conv1"),
      layers.Activation("tanh",name="tanh1"),
      layers.AveragePooling2D(pool_size=(2,2),name="avgpool1",strides=2),
      layers.Conv2D(16,(5,5),padding="valid",name="conv2"),
      layers.Activation("tanh",name="tanh2"),
      layers.AveragePooling2D(pool_size=(2,2),name="avgpool2",strides=2),
      layers.Flatten(name="flatten"),
      layers.Dense(120,name="dense1"),
      layers.Activation("tanh",name="tanh3"),
      layers.Dense(84,name="dense2"),
```

```
        layers.Activation("tanh",name="tanh4"),
        layers.Dense(num_classes,activation="softmax",name="output")
    ])
    return model

lenet = build_lenet5()
print(f"LeNet-5 Architecture Summary:")
lenet.summary()
print(f"\nManual Parameter Count Conv1(6filters,5x5,c_in=1)")
print(f"Formula: (kernel_h x kernel_w x c_in + 1_bias) x c_out")
k,c_in,c_out = 5,1,6
params_conv1 = (k*k*c_in+1)*c_out
print(f" = (5x5x1 + 1) x 6 = (25+1) = {params_conv1}")

total_params = sum(np.prod(v.shape)for v in lenet.trainable_variables)
print(f"\nTotal trainable parameters(from summary): {total_params}")
```

```
LeNet-5 Architecture Summary:
```

Model: "sequential"

Layer (type)	Output Shape	Param #
conv1 (Conv2D)	(None, 24, 24, 6)	156
tanh1 (Activation)	(None, 24, 24, 6)	0
avgpool1 (AveragePooling2D)	(None, 12, 12, 6)	0
conv2 (Conv2D)	(None, 8, 8, 16)	2,416
tanh2 (Activation)	(None, 8, 8, 16)	0
avgpool2 (AveragePooling2D)	(None, 4, 4, 16)	0
flatten (Flatten)	(None, 256)	0
dense1 (Dense)	(None, 120)	30,840
tanh3 (Activation)	(None, 120)	0
dense2 (Dense)	(None, 84)	10,164
tanh4 (Activation)	(None, 84)	0
output (Dense)	(None, 10)	850

Total params: 44,426 (173.54 KB)

Trainable params: 44,426 (173.54 KB)

Non-trainable params: 0 (0.00 B)

```
Manual Parameter Count Conv1(6filters,5x5,c_in=1)
Formula: (kernel_h x kernel_w x c_in + 1_bias) x c_out
 = (5x5x1 + 1) x 6 = (25+1) = 156

Total trainable parameters(from summary): 44426
```

5.4 *Design a Custom CNN for CIFAR-10*

[9]:
```python
def custom_cnn(input_shape=(32,32,3),num_classes=10):
  model = models.Sequential([
      layers.Input(shape=input_shape),
      layers.Conv2D(32,(3,3),padding="same",name="conv1"),
      layers.BatchNormalization(name="bn1"),
      layers.Activation("relu",name="relu1"),
      layers.MaxPooling2D((2,2),name="pool1"),
      layers.Dropout(0.25,name="drop1"),
      layers.Conv2D(64,(3,3),padding="same",name="conv2_1"),
      layers.BatchNormalization(name="bn2"),
      layers.Activation("relu",name="relu2"),
      layers.MaxPooling2D((2,2),name="pool2"),
      layers.Dropout(0.25,name="drop2"),
      layers.Conv2D(128,(3,3),padding="same",name="conv3_1"),
      layers.BatchNormalization(name="bn3"),
      layers.Activation("relu",name="relu3"),
      layers.MaxPooling2D((2,2),name="pool3"),
      layers.Dropout(0.25,name="drop3"),
      layers.GlobalAveragePooling2D(name="gap"),
      layers.Dense(256,name="dense1"),
      layers.Activation("relu",name="relu_dense"),
      layers.Dropout(0.5,name="drop_head"),
      layers.Dense(num_classes,activation="softmax",name="output"),
  ])

  return model
```

```python
cnn = custom_cnn()
print(f"Custom CNN Architecture Summary:")
cnn.summary()

total = sum(np.prod(v.shape) for v in cnn.trainable_variables)
print(f"\nTotal parameters: {total}")
```

```
Custom CNN Architecture Summary:
```

Model: "sequential_1"

Layer (type)	Output Shape	Param #
conv1 (Conv2D)	(None, 32, 32, 32)	896
bn1 (BatchNormalization)	(None, 32, 32, 32)	128
relu1 (Activation)	(None, 32, 32, 32)	0
pool1 (MaxPooling2D)	(None, 16, 16, 32)	0
drop1 (Dropout)	(None, 16, 16, 32)	0
conv2_1 (Conv2D)	(None, 16, 16, 64)	18,496
bn2 (BatchNormalization)	(None, 16, 16, 64)	256
relu2 (Activation)	(None, 16, 16, 64)	0
pool2 (MaxPooling2D)	(None, 8, 8, 64)	0
drop2 (Dropout)	(None, 8, 8, 64)	0
conv3_1 (Conv2D)	(None, 8, 8, 128)	73,856
bn3 (BatchNormalization)	(None, 8, 8, 128)	512
relu3 (Activation)	(None, 8, 8, 128)	0
pool3 (MaxPooling2D)	(None, 4, 4, 128)	0
drop3 (Dropout)	(None, 4, 4, 128)	0
gap (GlobalAveragePooling2D)	(None, 128)	0
dense1 (Dense)	(None, 256)	33,024

```
 relu_dense (Activation)          (None, 256)                        0

 drop_head (Dropout)              (None, 256)                        0

 output (Dense)                   (None, 10)                     2,570


 Total params: 129,738 (506.79 KB)

 Trainable params: 129,290 (505.04 KB)

 Non-trainable params: 448 (1.75 KB)


Total parameters: 129290
```

5.5 Analysis Questions

Q1. Compare the parameter efficiency of two stacked 3x3 Conv layers versus one 5x5 Conv layer on the same input with the same number of filters. Which uses fewer parameters? Show numerical proof and explain any other advantages of the smaller kernel approach.

Ans–
Stacking two 3x3 layers gives you the exact same 5x5 field of vision but allows you to sandwich an extra ReLU activation in between. This extra non-linearity makes your network more expressive and capable of modeling complex functions,due to which deep architectures like VGG and ResNet rely heavily on 3x3 convolutions.

Q2. What is the role of Batch Normalisation in a CNN? Where in the layer stack should it be placed (before or after activation), and why? Mention at least two empirical benefits it provides during training.

Ans–
Batch Norm acts as a stabilizer for your network. It takes the outputs of a layer and normalizes them across the mini-batch so they have a mean near 0 and a variance near 1. Placement is typically Conv->Batch Norm->Activation (ReLU). The benefist are: 1. It minimizes internal covariate shift. 2. Calculating statistics on random mini-batches injects a small noise into the network, which helps to prevent overfitting.

Q3. Your custom CNN has a GlobalAveragePooling layer before the Dense head. What does this layer do geometrically? What would happen to the parameter count and spatial information if you replaced it with Flatten?

Ans–
Geometry & parameters: Flatten converts the 3D feature map into a massive 1D vector, while GAP averages each channel into a single value, slashing parameter counts to prevent overfitting.

Information Trade-off: GAP discards spatial information, focusing only on the presence of features, making it ideal for classification but less effective for tasks requiring precise object localization.

6 *Training, Tuning & Regularisation (Task - 3)*

6.1 *FIRST TRAINING RUN (LeNet-5 on MNIST with SGD)*

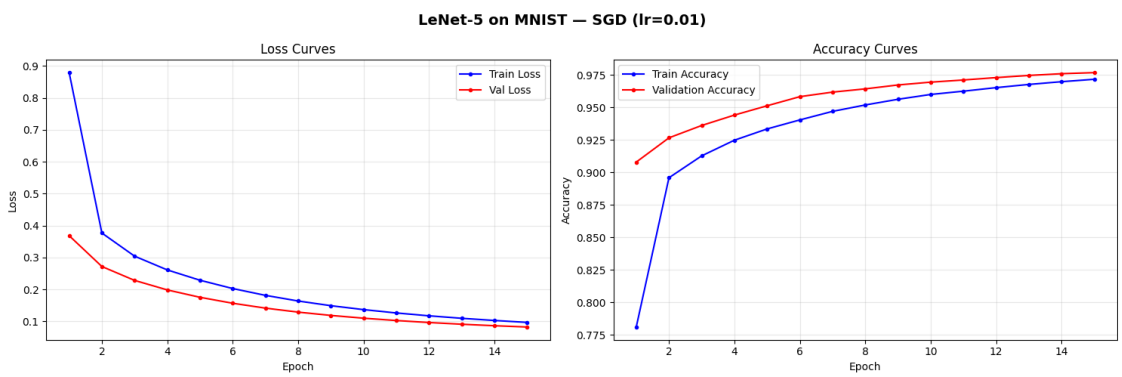
[10]:
```
from enum import verify
from tensorflow._api.v2.config import optimizer
def load_and_preprocess():
  (mx_tr,my_tr),(mx_te, my_te) = mnist.load_data()
  (cx_tr,cy_tr),(cx_te, cy_te) = cifar10.load_data()
  mx_tr = mx_tr.astype(np.float32) / 255.0
  mx_te = mx_te.astype(np.float32) / 255.0
  cx_tr = cx_tr.astype(np.float32) / 255.0
  cx_te = cx_te.astype(np.float32) / 255.0
  mx_tr = np.expand_dims(mx_tr, -1)
  mx_te = np.expand_dims(mx_te, -1)
  my_tr = keras.utils.to_categorical(my_tr, 10)
  my_te = keras.utils.to_categorical(my_te, 10)
  cy_tr = keras.utils.to_categorical(cy_tr, 10)
  cy_te = keras.utils.to_categorical(cy_te, 10)

  return (mx_tr, my_tr, mx_te, my_te), (cx_tr, cy_tr, cx_te, cy_te)

print(f"Loading the datasets...")
(mnist_train_x, mnist_train_y, mnist_test_x, mnist_test_y), \
(cifar10_train_x, cifar10_train_y, cifar10_test_x, cifar10_test_y) =␣
 ↪load_and_preprocess()
print(f"Loaded the dataset")

def build_cifar_cnn(dropout=True, batchnorm=True):
  model = models.Sequential(name=f"cifar_do{int(dropout)}_bn{int(batchnorm)}")
  model.add(layers.Input(shape=(32, 32, 3)))
  model.add(layers.Conv2D(32, (3,3), padding='same'))
  if batchnorm: model.add(layers.BatchNormalization())
  model.add(layers.Activation('relu'))
  model.add(layers.MaxPooling2D((2,2)))
  if dropout:  model.add(layers.Dropout(0.25))
  model.add(layers.Conv2D(64, (3,3), padding='same'))
  if batchnorm: model.add(layers.BatchNormalization())
  model.add(layers.Activation('relu'))
  model.add(layers.MaxPooling2D((2,2)))
  if dropout:  model.add(layers.Dropout(0.25))
  model.add(layers.Conv2D(128, (3,3), padding='same'))
  if batchnorm: model.add(layers.BatchNormalization())
  model.add(layers.Activation('relu'))
```

```
  model.add(layers.MaxPooling2D((2,2)))
  if dropout:  model.add(layers.Dropout(0.25))
  model.add(layers.GlobalAveragePooling2D())
  model.add(layers.Dense(256, activation='relu'))
  if dropout:  model.add(layers.Dropout(0.5))
  model.add(layers.Dense(10, activation='softmax'))

  return model

def plot_history(history, title, save_path):
  fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(15, 5))
  fig.suptitle(title, fontsize=14, fontweight='bold')
  epochs = range(1, len(history['loss']) + 1)
  ax1.plot(epochs, history['loss'],     'b-o', markersize=3, label='Train Loss')
  ax1.plot(epochs, history['val_loss'], 'r-o', markersize=3, label='Val Loss')
  ax1.set_xlabel('Epoch')
  ax1.set_ylabel('Loss')
  ax1.set_title('Loss Curves')
  ax1.legend()
  ax1.grid(True, alpha=0.3)
  ax2.plot(epochs, history['accuracy'],"b-o", markersize=3, label='Train␣
 ↪Accuracy')
  ax2.plot(epochs, history['val_accuracy'],"r-o", markersize=3,␣
 ↪label='Validation Accuracy')
  ax2.set_xlabel('Epoch')
  ax2.set_ylabel('Accuracy')
  ax2.set_title('Accuracy Curves')
  ax2.legend()
  ax2.grid(True, alpha=0.3)
  plt.tight_layout()
  plt.savefig(save_path, dpi=300)
  plt.show()

model_p1 = build_lenet5()
model_p1.compile(optimizer=keras.optimizers.SGD(learning_rate=0.
 ↪01),loss="categorical_crossentropy",metrics=["accuracy"])

history_p1 = model_p1.
 ↪fit(mnist_train_x,mnist_train_y,epochs=15,batch_size=64,validation_split=0.
 ↪1,verbose=1)

test_loss,test_acc = model_p1.evaluate(mnist_test_x,mnist_test_y,verbose=0)
print(f"Test accuarcy: {test_acc:.4f}")

val_losses = history_p1.history['val_loss']
overfit_epoch = None
for i in range(1, len(val_losses) - 1):
```

```
    if val_losses[i] > val_losses[i-1] and val_losses[i+1] > val_losses[i]:
        overfit_epoch = i + 1
        break

if overfit_epoch:
    print(f"Overfitting first observed at epoch {overfit_epoch}")
    print(f"Valdation loss at epoch {overfit_epoch-1}:␣
 ↪{val_losses[overfit_epoch-2]:.4f}")
    print(f"Validation loss at epoch {overfit_epoch}:␣
 ↪{val_losses[overfit_epoch-1]:.4f}")
else:
    print("No clear overfitting detected in 15 epochs.")

plot_history(history_p1.history, "LeNet-5 on MNIST - SGD (lr=0.01)",␣
 ↪"lenet_sgd_curves.png")
```

```
Loading the datasets…
Loaded the dataset
Epoch 1/15
844/844              8s 6ms/step -
accuracy: 0.7811 - loss: 0.8792 - val_accuracy: 0.9080 - val_loss: 0.3683
Epoch 2/15
844/844              2s 3ms/step -
accuracy: 0.8960 - loss: 0.3766 - val_accuracy: 0.9267 - val_loss: 0.2717
Epoch 3/15
844/844              2s 3ms/step -
accuracy: 0.9128 - loss: 0.3041 - val_accuracy: 0.9362 - val_loss: 0.2281
Epoch 4/15
844/844              2s 3ms/step -
accuracy: 0.9248 - loss: 0.2609 - val_accuracy: 0.9442 - val_loss: 0.1983
Epoch 5/15
844/844              2s 3ms/step -
accuracy: 0.9335 - loss: 0.2286 - val_accuracy: 0.9513 - val_loss: 0.1751
Epoch 6/15
844/844              3s 3ms/step -
accuracy: 0.9404 - loss: 0.2026 - val_accuracy: 0.9583 - val_loss: 0.1565
Epoch 7/15
844/844              5s 3ms/step -
accuracy: 0.9470 - loss: 0.1812 - val_accuracy: 0.9618 - val_loss: 0.1413
Epoch 8/15
844/844              2s 3ms/step -
accuracy: 0.9519 - loss: 0.1636 - val_accuracy: 0.9643 - val_loss: 0.1287
Epoch 9/15
844/844              2s 3ms/step -
accuracy: 0.9563 - loss: 0.1489 - val_accuracy: 0.9673 - val_loss: 0.1184
Epoch 10/15
844/844              3s 3ms/step -
```

```
accuracy: 0.9600 - loss: 0.1366 - val_accuracy: 0.9695 - val_loss: 0.1097
Epoch 11/15
844/844              3s 3ms/step -
accuracy: 0.9626 - loss: 0.1261 - val_accuracy: 0.9712 - val_loss: 0.1024
Epoch 12/15
844/844              2s 3ms/step -
accuracy: 0.9653 - loss: 0.1171 - val_accuracy: 0.9730 - val_loss: 0.0962
Epoch 13/15
844/844              2s 3ms/step -
accuracy: 0.9677 - loss: 0.1094 - val_accuracy: 0.9747 - val_loss: 0.0908
Epoch 14/15
844/844              2s 3ms/step -
accuracy: 0.9699 - loss: 0.1026 - val_accuracy: 0.9760 - val_loss: 0.0861
Epoch 15/15
844/844              3s 3ms/step -
accuracy: 0.9717 - loss: 0.0966 - val_accuracy: 0.9768 - val_loss: 0.0820
Test accuarcy: 0.9728
No clear overfitting detected in 15 epochs.
```

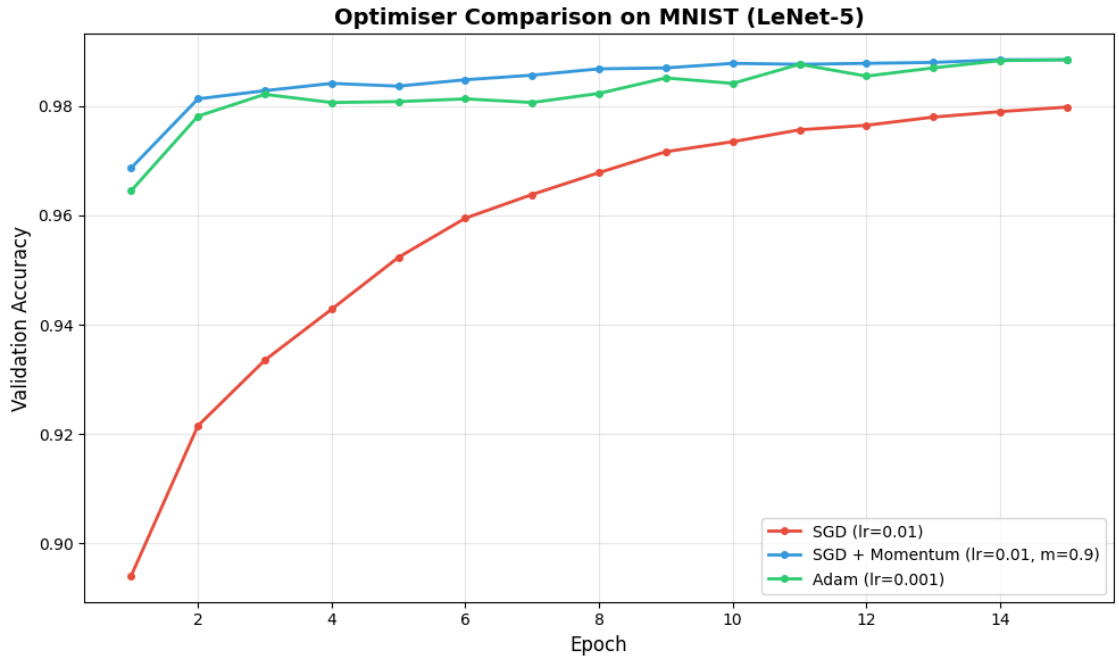


6.2 *OPTIMISER COMPARISON*

[11]:
```
optimisers = {
  'SGD (lr=0.01)':keras.optimizers.SGD(learning_rate=0.01),
  'SGD + Momentum (lr=0.01, m=0.9)':keras.optimizers.SGD(learning_rate=0.01,␣
 ↪momentum=0.9),
  'Adam (lr=0.001)':keras.optimizers.Adam(learning_rate=0.001),
}
hist_opt = {}
for name, opt in optimisers.items():
  print(f"Training with: {name}")
  np.random.seed(42)
  tf.random.set_seed(42)
  m = build_lenet5()
```

```
  m.compile(optimizer=opt, loss='categorical_crossentropy',␣
 ↪metrics=['accuracy'])
  h = m.fit(
      mnist_train_x, mnist_train_y,
      epochs=15, batch_size=64,
      validation_split=0.1,
      verbose=0
    )
  hist_opt[name] = h.history
  print(f"Final validation accuracy: {h.history['val_accuracy'][-1]:.4f}")

fig, ax = plt.subplots(figsize=(10, 6))
colors = ['#e74c3c', '#3498db', '#2ecc71']
for (name, hist), color in zip(hist_opt.items(), colors):
    ax.plot(range(1, 16), hist['val_accuracy'], '-o', markersize=4,
            label=name, color=color, linewidth=2)
ax.set_xlabel('Epoch', fontsize=12)
ax.set_ylabel('Validation Accuracy', fontsize=12)
ax.set_title('Optimiser Comparison on MNIST (LeNet-5)', fontsize=14,␣
 ↪fontweight='bold')
ax.legend(fontsize=10)
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig('optimiser_comparison.png', dpi=300)
plt.show()
```

```
Training with: SGD (lr=0.01)
Final validation accuracy: 0.9798
Training with: SGD + Momentum (lr=0.01, m=0.9)
Final validation accuracy: 0.9885
Training with: Adam (lr=0.001)
Final validation accuracy: 0.9885
```



6.3 *LEARNING RATE & BATCH SIZE GRID SEARCH*

[12]:
```
learning_rates = [0.1, 0.01, 0.001]
batch_sizes    = [32, 128]
results_grid   = {}

for lr in learning_rates:
  for bs in batch_sizes:
      print(f"Training: lr={lr}, batch_size={bs}")
      np.random.seed(42)
      tf.random.set_seed(42)
      m = build_cifar_cnn(dropout=True, batchnorm=True)
      m.compile(
          optimizer=keras.optimizers.Adam(learning_rate=lr),
          loss='categorical_crossentropy',
          metrics=['accuracy']
      )
      h = m.fit(
        cifar10_train_x, cifar10_train_y,
        epochs=10,
        batch_size=bs,
        validation_split=0.1,
        verbose=0
      )
      val_acc = h.history['val_accuracy'][-1]
```

```
        results_grid[(lr, bs)] = val_acc
        print(f"Validation accuracy: {val_acc:.4f}")

print("\nGrid Search Results Table:")
print(f"{'LR/Batch':>10} | {'BS=32':>8} | {'BS=128':>8}")
best_combo = max(results_grid, key=results_grid.get)
for lr in learning_rates:
    row = f"{lr:>10} | "
    for bs in batch_sizes:
        v = results_grid[(lr, bs)]
        marker = "  " if (lr, bs) == best_combo else ""
        row += f"{v:.4f}{marker:>3} | "
    print(row)
print(f"\nBest combination: lr={best_combo[0]}, batch_size={best_combo[1]}:␣
↪{results_grid[best_combo]:.4f}")
```

```
Training: lr=0.1, batch_size=32
Validation accuracy: 0.0970
Training: lr=0.1, batch_size=128
Validation accuracy: 0.0970
Training: lr=0.01, batch_size=32
Validation accuracy: 0.6256
Training: lr=0.01, batch_size=128
Validation accuracy: 0.6048
Training: lr=0.001, batch_size=32
Validation accuracy: 0.6370
Training: lr=0.001, batch_size=128
Validation accuracy: 0.6326

Grid Search Results Table:
  LR/Batch |    BS=32 |   BS=128
       0.1 | 0.0970    | 0.0970    |
      0.01 | 0.6256    | 0.6048    |
     0.001 | 0.6370    | 0.6326    |

Best combination: lr=0.001, batch_size=32: 0.6370
```

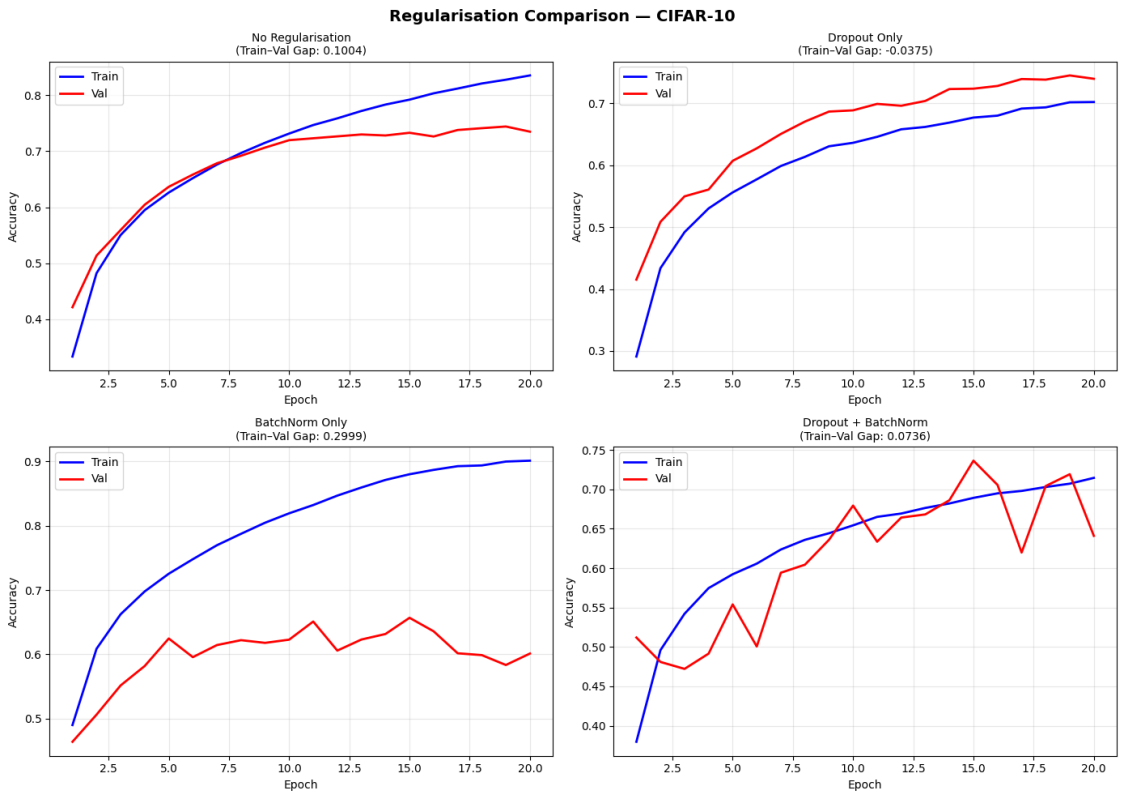
6.4 *REGULARISATION EXPERIMENT*

[13]:
```
variants = {
  'No Regularisation':build_cifar_cnn(dropout=False, batchnorm=False),
  'Dropout Only':build_cifar_cnn(dropout=True,  batchnorm=False),
  'BatchNorm Only':build_cifar_cnn(dropout=False, batchnorm=True),
  'Dropout + BatchNorm':build_cifar_cnn(dropout=True,  batchnorm=True),
}

hist_reg = {}
```

```python
for name, model in variants.items():
    print(f"\nTraining: {name}")
    np.random.seed(42)
    tf.random.set_seed(42)
    model.compile(
        optimizer=keras.optimizers.Adam(0.001),
        loss='categorical_crossentropy',
        metrics=['accuracy']
    )
    h = model.fit(
        cifar10_train_x,cifar10_train_y,
        epochs=20,
        batch_size=64,
        validation_split=0.1,
        verbose=0
    )
    hist_reg[name] = h.history
    train_acc = h.history['accuracy'][-1]
    val_acc = h.history['val_accuracy'][-1]
    gap = train_acc - val_acc
    print(f"Train accuracy: {train_acc:.4f} | Val acc: {val_acc:.4f} | Gap:␣
 ↪{gap:.4f}")

fig, axes = plt.subplots(2, 2, figsize=(14, 10))
fig.suptitle('Regularisation Comparison -␣
 ↪CIFAR-10',fontsize=14,fontweight='bold')

for ax, (name, hist) in zip(axes.flatten(), hist_reg.items()):
  eps = range(1, len(hist['accuracy']) + 1)
  ax.plot(eps,hist['accuracy'],'b-',label='Train',linewidth=2)
  ax.plot(eps,hist['val_accuracy'],'r-',label='Val',linewidth=2)
  gap = hist['accuracy'][-1] - hist['val_accuracy'][-1]
  ax.set_title(f'{name}\n(Train-Val Gap: {gap:.4f})', fontsize=10)
  ax.set_xlabel('Epoch')
  ax.set_ylabel('Accuracy')
  ax.legend()
  ax.grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig('regularisation_comparison.png', dpi=300)
plt.show()

print("\nRegularisation Summary Table")
print(f"{'Variant':<25} | {'Train Acc':>9} | {'Val Acc':>8} | {'Gap':>8}")
for name, hist in hist_reg.items():
  tr = hist['accuracy'][-1]
  va = hist['val_accuracy'][-1]
```

```python
print(f"{name:<25} | {tr:>9.4f} | {va:>8.4f} | {tr-va:>8.4f}")
```

```
Training: No Regularisation
Train accuracy: 0.8352 | Val acc: 0.7348 | Gap: 0.1004

Training: Dropout Only
Train accuracy: 0.7023 | Val acc: 0.7398 | Gap: -0.0375

Training: BatchNorm Only
Train accuracy: 0.9011 | Val acc: 0.6012 | Gap: 0.2999

Training: Dropout + BatchNorm
Train accuracy: 0.7146 | Val acc: 0.6410 | Gap: 0.0736
```



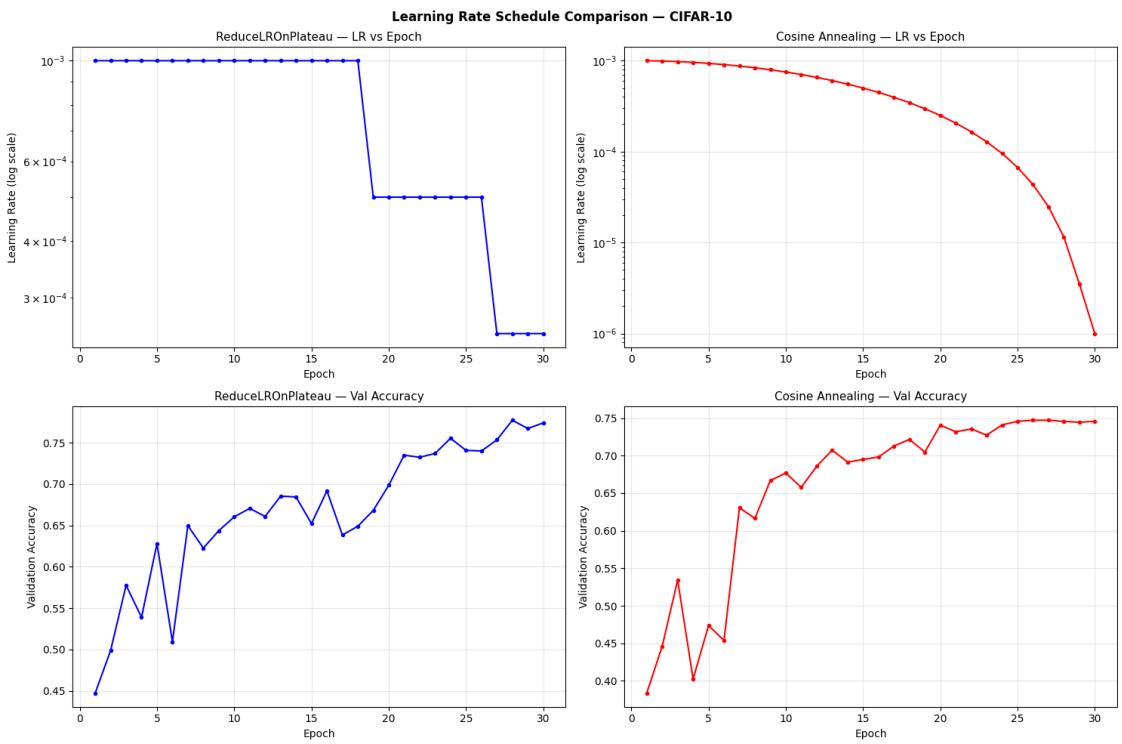

```
Regularisation Summary Table
Variant                   | Train Acc |  Val Acc |      Gap
No Regularisation         |    0.8352 |   0.7348 |   0.1004
Dropout Only              |    0.7023 |   0.7398 |  -0.0375
BatchNorm Only            |    0.9011 |   0.6012 |   0.2999
Dropout + BatchNorm       |    0.7146 |   0.6410 |   0.0736
```

[14]:
```
EPOCHS = 30
INIT_LR = 0.001

print("Training with ReduceLROnPlateau")
np.random.seed(42); tf.random.set_seed(42)
model_plateau = build_cifar_cnn(dropout=True, batchnorm=True)
model_plateau.compile(
  optimizer=keras.optimizers.Adam(INIT_LR),
  loss='categorical_crossentropy',
  metrics=['accuracy']
)

lr_tracker_plateau = []
class LRLogger(callbacks.Callback):
  def on_epoch_end(self, epoch, logs=None):
      lr_tracker_plateau.append(float(self.model.optimizer.learning_rate))

h_plateau = model_plateau.fit(
  cifar10_train_x, cifar10_train_y,
  epochs=EPOCHS, batch_size=64,
  validation_split=0.1,
  callbacks=[
      callbacks.ReduceLROnPlateau(
          monitor='val_loss', factor=0.5, patience=3,
          min_lr=1e-6, verbose=1
      ),
      LRLogger()
  ],
  verbose=0)
print("\nTraining with Cosine Annealing")
np.random.seed(42); tf.random.set_seed(42)

cosine_schedule = keras.optimizers.schedules.CosineDecay(
    initial_learning_rate=INIT_LR,
    decay_steps=EPOCHS * (len(cifar10_train_x) * 0.9 // 64),
    alpha=1e-6/INIT_LR
)

model_cosine = build_cifar_cnn(dropout=True, batchnorm=True)
model_cosine.compile(
  optimizer=keras.optimizers.Adam(cosine_schedule),
  loss='categorical_crossentropy',
  metrics=['accuracy']
)

lr_tracker_cosine = []
class LRLoggerCosine(callbacks.Callback):
```

```
  def on_epoch_end(self, epoch, logs=None):
      opt = self.model.optimizer
      if hasattr(opt.learning_rate, '__call__'):
          step = opt.iterations.numpy()
          lr = float(opt.learning_rate(step))
      else:
          lr = float(opt.learning_rate)
      lr_tracker_cosine.append(lr)

h_cosine = model_cosine.fit(
  cifar10_train_x, cifar10_train_y,
  epochs=EPOCHS, batch_size=64,
  validation_split=0.1,
  callbacks=[LRLoggerCosine()],
  verbose=0
)

fig, axes = plt.subplots(2, 2, figsize=(15, 10))
fig.suptitle('Learning Rate Schedule Comparison - CIFAR-10', fontsize=12,␣
 ↪fontweight='bold')
eps = range(1, EPOCHS + 1)

axes[0,0].semilogy(eps, lr_tracker_plateau[:EPOCHS],'b-o', markersize=3)
axes[0,0].set_title('ReduceLROnPlateau - LR vs Epoch', fontsize=11)
axes[0,0].set_xlabel('Epoch')
axes[0,0].set_ylabel('Learning Rate (log scale)')
axes[0,0].grid(True, alpha=0.3)

axes[0,1].semilogy(eps, lr_tracker_cosine[:EPOCHS],'r-o', markersize=3)
axes[0,1].set_title('Cosine Annealing - LR vs Epoch', fontsize=11)
axes[0,1].set_xlabel('Epoch')
axes[0,1].set_ylabel('Learning Rate (log scale)')
axes[0,1].grid(True, alpha=0.3)

axes[1,0].plot(eps, h_plateau.history['val_accuracy'][:EPOCHS], 'b-o',␣
 ↪markersize=3)
axes[1,0].set_title('ReduceLROnPlateau - Val Accuracy', fontsize=11)
axes[1,0].set_xlabel('Epoch')
axes[1,0].set_ylabel('Validation Accuracy')
axes[1,0].grid(True, alpha=0.3)

axes[1,1].plot(eps, h_cosine.history['val_accuracy'][:EPOCHS], 'r-o',␣
 ↪markersize=3)
axes[1,1].set_title('Cosine Annealing - Val Accuracy', fontsize=11)
axes[1,1].set_xlabel('Epoch')
axes[1,1].set_ylabel('Validation Accuracy')
axes[1,1].grid(True, alpha=0.3)
```

```
plt.tight_layout()
plt.savefig('lr_schedule_comparison.png', dpi=300)
plt.show()
best_plateau = max(h_plateau.history['val_accuracy'])
best_cosine  = max(h_cosine.history['val_accuracy'])
print(f"\nBest val accuracy - ReduceLROnPlateau: {best_plateau:.4f}")
print(f"Best val accuracy - Cosine Annealing:  {best_cosine:.4f}")
```

```
Training with ReduceLROnPlateau

Epoch 19: ReduceLROnPlateau reducing learning rate to 0.00050000000237487257.

Epoch 27: ReduceLROnPlateau reducing learning rate to 0.00025000000118743628.

Training with Cosine Annealing
```




```
Best val accuracy - ReduceLROnPlateau: 0.7772
Best val accuracy - Cosine Annealing:  0.7472
```

6.5 Analysis Questions

Q1. Explain why a very high learning rate (e.g., 1.0) can cause training loss to diverge or oscillate rather than converge. Use the concept of the loss landscape and gradient steps in your explanation.

Ans–
Normal learning rate: Takes small, steady steps to gently converge at the minimum.

Too high learning rate (e.g., lr=1.0): Takes massive steps that completely overshoot the valley. Instead of settling at the bottom, the loss bounces back and forth (oscillates) or diverges entirely.

The Math: Because updates are calculated as weight_new = weight - (lr * gradient), a high learning rate multiplies the gradient too much. This throws the model into completely wrong weight spaces, which is dangerous in narrow valleys or saddle points.

Q2. Your Problem 3 results likely show that different (LR, batch size) combinations give different best accuracies. From your data: which combination worked best and which worked worst? Propose a hypothesis explaining the pattern you observed.

Ans–
Small Batch + Low LR (Optimal): Small batches introduce gradient noise, allowing the optimizer to escape sharp local minima and find flatter,regions that generalize better. A low learning rate ensures these updates remain stable.

Large Batch + High LR (Suboptimal): Large batches provide smooth, accurate gradients that can trap the model in narrow, sharp minima which perform poorly on new data. A high learning rate significantly increases the risk of overshooting the optimal solution entirely.

Q3. Dropout is disabled at inference (test) time. Why? If a network has Dropout(0.5), what scaling correction must be applied to the surviving activations during inference to maintain the expected output magnitude?

Ans–
Deployed models need deterministic predictions. Leaving Dropout on would yield random, inconsistent outputs for the exact same input every time you run it. Because 100% of neurons fire during inference (compared to 50% during training with a 0.5 rate), the total output magnitude doubles. To compensate, test-time activations must be multiplied by (1 - rate), which is 0.5.

Q4. Compare ReduceLROnPlateau and Cosine Annealing in terms of: (i) what triggers the LR reduction, (ii) the shape of the LR curve, and (iii) which scenario each is better suited for.

Ans–
ReduceLROnPlateau is a reactive scheduler that drops the learning rate in a sharp "staircase" pattern only when the validation loss plateaus. It is ideal where the training dynamics are unknown. In contrast, Cosine Annealing is proactive, continuously decaying the learning rate along a smooth S-curve based on a fixed schedule. It is best for fine-tuning where the total number of epochs is known and you need to get the maximum final performance.

7 *Visualisation & Interpretability (Task - 4)*

7.1 *Visualise Learned Filters*

[15]:
```
cifar10_test_y_raw = np.argmax(cifar10_test_y, axis=1)
print(f"Training the CIFAR10 model")
model = build_cifar_cnn(dropout=True, batchnorm=True)
model.compile(
  optimizer=keras.optimizers.Adam(0.001),
  loss='categorical_crossentropy',
  metrics=['accuracy']
)

model.fit(
  cifar10_train_x, cifar10_train_y,
  epochs=20, batch_size=64,
  validation_split=0.1,
  verbose=1
)
print(f"Model has been tarined")
test_loss,test_acc = model.evaluate(cifar10_test_x,cifar10_test_y,verbose=0)
print(f"Test accuarcy: {test_acc:.4f}")

conv1_layer = [layer for layer in model.layers if isinstance(layer, layers.
 ↪Conv2D)][0]
conv1_w = conv1_layer.get_weights()[0]

n_filters = conv1_w.shape[3]
n_cols = 8
n_rows = (n_filters+n_cols-1)//n_cols

fig,axes = plt.subplots(n_rows,n_cols,figsize=(15,8))
fig.suptitle('Conv1 Learned Filters(normalized)',fontsize=14,fontweight='bold')

for i in range(n_rows*n_cols):
  ax = axes[i//n_cols,i%n_cols]
  if(i<n_filters):
    filt = conv1_w[:,:,:,i]
    filt_norm = (filt-np.min(filt))/(np.max(filt)-np.min(filt)+1e-8)
    ax.imshow(filt_norm)
    ax.set_title(f"F{i}",fontsize=7)
  ax.axis("off")

plt.tight_layout()
plt.savefig('conv1_filters.png',dpi=300)
plt.show()
```

```
Training the CIFAR10 model
```

```
Epoch 1/20
704/704              16s 13ms/step -
accuracy: 0.3851 - loss: 1.6652 - val_accuracy: 0.3764 - val_loss: 1.6864
Epoch 2/20
704/704              4s 5ms/step -
accuracy: 0.4993 - loss: 1.3772 - val_accuracy: 0.4248 - val_loss: 1.5873
Epoch 3/20
704/704              4s 6ms/step -
accuracy: 0.5405 - loss: 1.2670 - val_accuracy: 0.5368 - val_loss: 1.2367
Epoch 4/20
704/704              4s 5ms/step -
accuracy: 0.5694 - loss: 1.1928 - val_accuracy: 0.5362 - val_loss: 1.2727
Epoch 5/20
704/704              4s 5ms/step -
accuracy: 0.5912 - loss: 1.1453 - val_accuracy: 0.5578 - val_loss: 1.2272
Epoch 6/20
704/704              4s 6ms/step -
accuracy: 0.6031 - loss: 1.1065 - val_accuracy: 0.5412 - val_loss: 1.1925
Epoch 7/20
704/704              4s 5ms/step -
accuracy: 0.6220 - loss: 1.0659 - val_accuracy: 0.6428 - val_loss: 0.9842
Epoch 8/20
704/704              4s 5ms/step -
accuracy: 0.6323 - loss: 1.0310 - val_accuracy: 0.5840 - val_loss: 1.1436
Epoch 9/20
704/704              4s 6ms/step -
accuracy: 0.6430 - loss: 1.0043 - val_accuracy: 0.6434 - val_loss: 1.0189
Epoch 10/20
704/704              4s 5ms/step -
accuracy: 0.6539 - loss: 0.9827 - val_accuracy: 0.6594 - val_loss: 0.9376
Epoch 11/20
704/704              4s 5ms/step -
accuracy: 0.6602 - loss: 0.9622 - val_accuracy: 0.6930 - val_loss: 0.8516
Epoch 12/20
704/704              4s 5ms/step -
accuracy: 0.6661 - loss: 0.9466 - val_accuracy: 0.6928 - val_loss: 0.8614
Epoch 13/20
704/704              4s 6ms/step -
accuracy: 0.6743 - loss: 0.9224 - val_accuracy: 0.7076 - val_loss: 0.8223
Epoch 14/20
704/704              4s 5ms/step -
accuracy: 0.6800 - loss: 0.9041 - val_accuracy: 0.7030 - val_loss: 0.8470
Epoch 15/20
704/704              4s 5ms/step -
accuracy: 0.6866 - loss: 0.8917 - val_accuracy: 0.6318 - val_loss: 1.0301
Epoch 16/20
704/704              4s 6ms/step -
accuracy: 0.6905 - loss: 0.8777 - val_accuracy: 0.6446 - val_loss: 0.9822
```

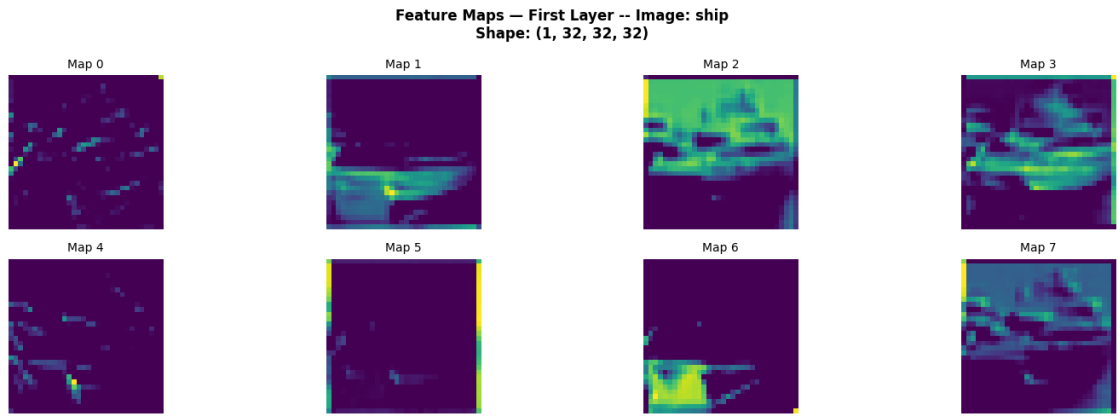
```
Epoch 17/20
704/704              4s 5ms/step -
accuracy: 0.6980 - loss: 0.8644 - val_accuracy: 0.6926 - val_loss: 0.8814
Epoch 18/20
704/704              4s 5ms/step -
accuracy: 0.7019 - loss: 0.8558 - val_accuracy: 0.6886 - val_loss: 0.8697
Epoch 19/20
704/704              4s 6ms/step -
accuracy: 0.7064 - loss: 0.8428 - val_accuracy: 0.7256 - val_loss: 0.7885
Epoch 20/20
704/704              4s 5ms/step -
accuracy: 0.7122 - loss: 0.8264 - val_accuracy: 0.7206 - val_loss: 0.8193
Model has been tarined
Test accuarcy: 0.7082
```

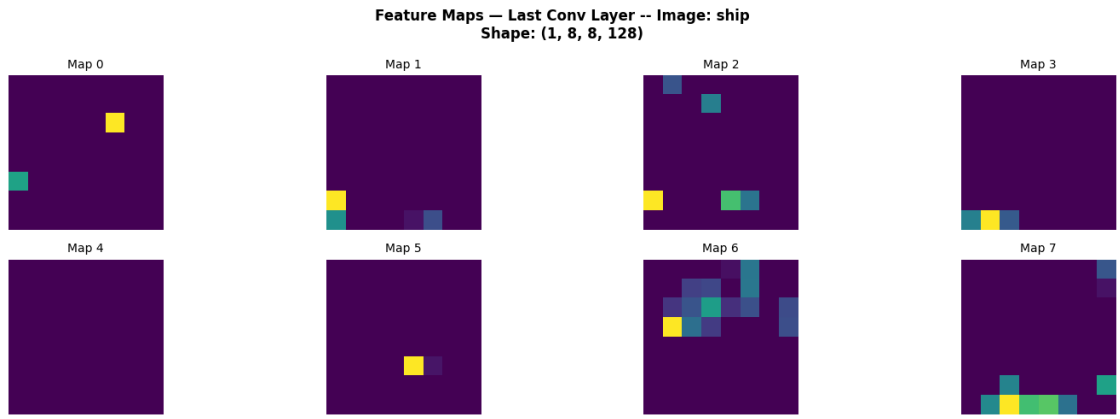
7.2 *Intermediate Feature Maps*

[16]:
```
act_layers = [layer.name for layer in model.layers if isinstance(layer, layers.
↪Activation)]
layer_names = [act_layers[0], act_layers[-1]]

feature_models = {}
for name in layer_names:
  feature_models[name] = keras.Model(
    inputs=model.inputs,
    outputs=model.get_layer(name).output
    )
```

```
pred = model.predict(cifar10_test_x[:200], verbose=0)
pred_classes = np.argmax(pred, axis=1)
true_classes = cifar10_test_y_raw[:200]
correct_id  = np.where(pred_classes == true_classes)[0][0]

test_img = cifar10_test_x[correct_id]
true_label = cifar10_class_names[true_classes[correct_id]]
pred_label  = cifar10_class_names[pred_classes[correct_id]]
print(f"Selected image: true={true_label}, predicted={pred_label}")

img_input = np.expand_dims(test_img, axis=0)

def plot_feature_maps(fmaps,title,save_path=None,n_maps=8):
  fig, axes = plt.subplots(2, n_maps // 2, figsize=(16, 5))
  fig.suptitle(f'{title}\nShape: {fmaps.shape}', fontsize=12, fontweight='bold')

  for i in range(n_maps):
      ax = axes[i//(n_maps// 2),i%(n_maps//2)]
      fmap = fmaps[0, :, :, i]
      fmap_norm = (fmap-fmap.min()) / (fmap.max() - fmap.min() + 1e-8)
      ax.imshow(fmap_norm,cmap='viridis')
      ax.set_title(f'Map {i}',fontsize=10)
      ax.axis('off')

  plt.tight_layout()
  if save_path:
    plt.savefig(save_path,dpi=300)
  plt.show()

fmaps_layer1 = feature_models[layer_names[0]].predict(img_input, verbose=0)
plot_feature_maps(fmaps_layer1,f'Feature Maps - First Layer -- Image:␣
 ↪{true_label}','fmaps_layer1.png',n_maps=8)

fmaps_last = feature_models[layer_names[1]].predict(img_input, verbose=0)
plot_feature_maps(fmaps_last,f'Feature Maps - Last Conv Layer -- Image:␣
 ↪{true_label}','fmaps_last.png',n_maps=8)
```

```
Selected image: true=ship, predicted=ship
```



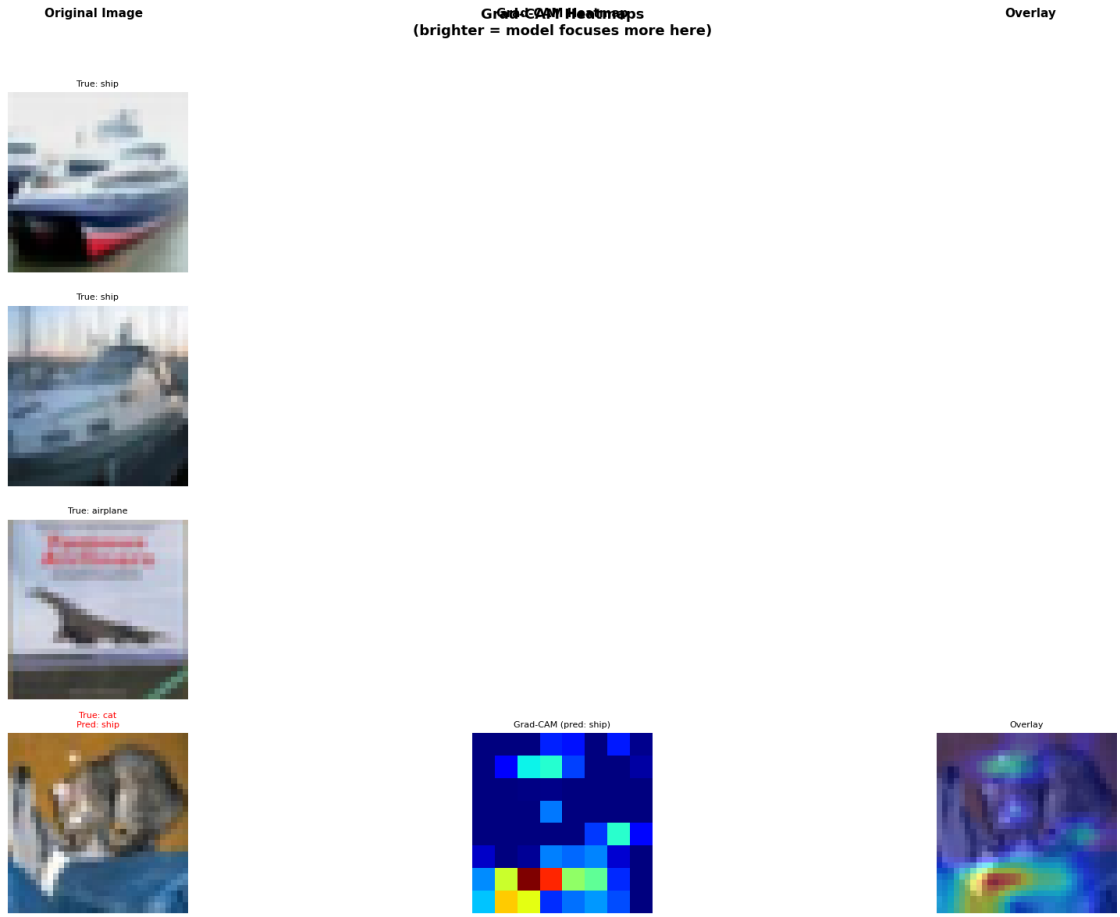


7.3 *Grad-CAM Implementation (from scratch)*

[17]:
```
last_conv_layer_name = [layer.name for layer in model.layers if␣
 ↪isinstance(layer, layers.Conv2D)][-1]

grad_input = keras.Input(shape=(32, 32, 3))
x = grad_input
conv_output = None
for layer in model.layers:
 x = layer(x)
 if layer.name == last_conv_layer_name:
  conv_output = x
grad_model = keras.Model(inputs=grad_input, outputs=[conv_output, x])

def compute_gradcam(img_array, class_idx):
 with tf.GradientTape() as tape:
  img_tensor = tf.cast(img_array, tf.float32)
```

```
  conv_outputs, pred = grad_model(img_tensor)
  tape.watch(conv_outputs)
  class_score = pred[:, class_idx]

 grads = tape.gradient(class_score, conv_outputs)
 pooled_grads = tf.reduce_mean(grads, axis=(0, 1, 2))
 conv_outputs_np = conv_outputs[0].numpy()
 pooled_grads_np = pooled_grads.numpy()

 for k in range(pooled_grads_np.shape[0]):
  conv_outputs_np[:, :, k] *= pooled_grads_np[k]

 heatmap = np.mean(conv_outputs_np, axis=-1)
 heatmap = np.maximum(heatmap, 0)

 if heatmap.max() > 0:
  heatmap /= heatmap.max()
 return heatmap

def overlay_heatmap(image, heatmap, alpha=0.5, colormap=cm.jet):
 H, W = image.shape[:2]
 h, w = heatmap.shape
 hmap_pil = PILImage.fromarray((heatmap * 255).astype(np.uint8))
 hmap_pil = hmap_pil.resize((W, H), PILImage.BILINEAR)
 heatmap_resized = np.array(hmap_pil).astype(np.float32) / 255.0
 heatmap_coloured = colormap(heatmap_resized)[:, :, :3]
 overlaid = (1 - alpha) * image + alpha * heatmap_coloured
 return np.clip(overlaid, 0, 1)

all_preds = model.predict(cifar10_test_x, verbose=0)
pred_all = np.argmax(all_preds, axis=1)

correct_indices  = np.where(pred_all == cifar10_test_y_raw)[0]
misclassified_idxs = np.where(pred_all != cifar10_test_y_raw)[0]

correct_3  = correct_indices[:3]
misclass_1 = misclassified_idxs[0]

fig = plt.figure(figsize=(18, 12))
fig.suptitle('Grad-CAM Heatmaps\n(brighter = model focuses more here)',␣
 ↪fontsize=13, fontweight='bold')

rows, cols = 4, 3
col_labels = ['Original Image', 'Grad-CAM Heatmap', 'Overlay']
for col, label in enumerate(col_labels):
 fig.text((col + 0.5) / cols, 0.97, label, ha='center', fontsize=11,␣
 ↪fontweight='bold')
```

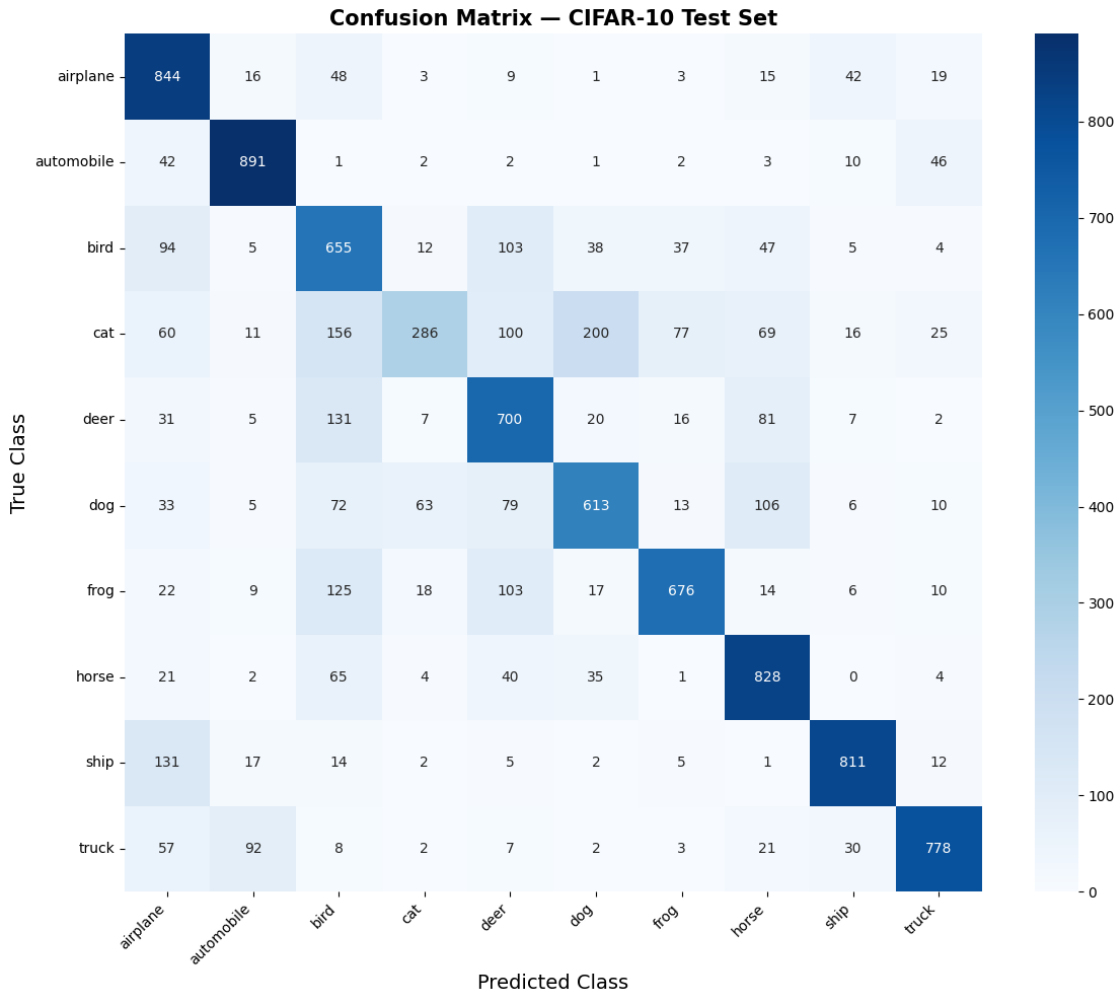
```
for row, idx in enumerate(list(correct_3) + [misclass_1]):
 img  = cifar10_test_x[idx]
 true_c = cifar10_test_y_raw[idx]
 pred_c = pred_all[idx]
 img_input = np.expand_dims(img, 0)
 heatmap_pred = compute_gradcam(img_input, pred_c)

 ax1 = fig.add_subplot(rows, cols, row * cols + 1)
 ax1.imshow(img)
 if row == 3:
  ax1.set_title(f'True: {cifar10_class_names[true_c]}\nPred:␣
 ↪{cifar10_class_names[pred_c]}', fontsize=8, color='red')
 else:
  ax1.set_title(f'True: {cifar10_class_names[true_c]}', fontsize=8)
 ax1.axis('off')

ax2 = fig.add_subplot(rows, cols, row * cols + 2)
ax2.imshow(heatmap_pred, cmap='jet')
ax2.set_title(f'Grad-CAM (pred: {cifar10_class_names[pred_c]})', fontsize=8)
ax2.axis('off')

ax3 = fig.add_subplot(rows, cols, row * cols + 3)
overlay = overlay_heatmap(img, heatmap_pred)
ax3.imshow(overlay)
ax3.set_title('Overlay', fontsize=8)
ax3.axis('off')

plt.tight_layout(rect=[0, 0, 1, 0.96])
plt.savefig('gradcam_results.png', dpi=300)
plt.show()
```



7.4 *Confusion Matrix & Classification Report*

[18]:
```
cm_result = confusion_matrix(cifar10_test_y_raw, pred_all)

fig, ax = plt.subplots(figsize=(12, 10))
sns.heatmap(cm_result, annot=True, fmt='d', cmap='Blues',
      xticklabels=cifar10_class_names,
      yticklabels=cifar10_class_names,
      ax=ax)
ax.set_xlabel('Predicted Class', fontsize=14)
ax.set_ylabel('True Class', fontsize=14)
ax.set_title('Confusion Matrix - CIFAR-10 Test Set', fontsize=15,␣
 ↪fontweight='bold')
plt.xticks(rotation=45, ha='right')
plt.yticks(rotation=0)
plt.tight_layout()
plt.savefig('confusion_matrix.png', dpi=300)
plt.show()
```

```
print("\nClassification Report->")
print(classification_report(cifar10_test_y_raw, pred_all,␣
↪target_names=cifar10_class_names))

f1_per_class = f1_score(cifar10_test_y_raw, pred_all, average=None)
best_class  = cifar10_class_names[np.argmax(f1_per_class)]
worst_class = cifar10_class_names[np.argmin(f1_per_class)]

print(f"\nHighest F1: {best_class} ({f1_per_class.max():.4f})")
print(f"Lowest  F1: {worst_class} ({f1_per_class.min():.4f})")

cm_no_diag = cm_result.copy()
np.fill_diagonal(cm_no_diag, 0)
flat_idx = np.argsort(cm_no_diag.flatten())[::-1]
pair1_i, pair1_j = flat_idx[0] // 10, flat_idx[0] % 10

print(f"\nMost confused pair: {cifar10_class_names[pair1_i]} → predicted as␣
↪{cifar10_class_names[pair1_j]}: {cm_no_diag[pair1_i, pair1_j]} errors")

misclass_pair = np.where((cifar10_test_y_raw == pair1_i) & (pred_all ==␣
↪pair1_j))[0]
n_show = min(5,len(misclass_pair))

fig,axes = plt.subplots(1, n_show, figsize=(3 * n_show, 4))
fig.suptitle(f'Misclassified:␣
↪True={cifar10_class_names[pair1_i]},Predicted={cifar10_class_names[pair1_j]}',fontsize=12,f

for i in range(n_show):
  idx = misclass_pair[i]
  axes[i].imshow(cifar10_test_x[idx])
  axes[i].set_title(f'Pred:␣
↪{cifar10_class_names[pred_all[idx]]}',fontsize=10,color='red')
  axes[i].axis('off')

plt.tight_layout()
plt.savefig('misclassified_examples.png', dpi=300)
plt.show()
```




```
Classification Report->
              precision    recall  f1-score   support

    airplane       0.63      0.84      0.72      1000
  automobile       0.85      0.89      0.87      1000
        bird       0.51      0.66      0.58      1000
         cat       0.72      0.29      0.41      1000
        deer       0.61      0.70      0.65      1000
         dog       0.66      0.61      0.64      1000
        frog       0.81      0.68      0.74      1000
       horse       0.70      0.83      0.76      1000
        ship       0.87      0.81      0.84      1000
       truck       0.85      0.78      0.81      1000

    accuracy                           0.71     10000
   macro avg       0.72      0.71      0.70     10000
```

```
weighted avg       0.72      0.71      0.70     10000


Highest F1: automobile (0.8680)
Lowest  F1: cat (0.4089)

Most confused pair: cat → predicted as dog: 200 errors
```

7.5 Analysis Questions

Q1. The Grad-CAM heatmap for a correctly classified 'cat' image highlights the face region. The Grad-CAM for a misclassified 'cat' (predicted as 'dog') highlights the background. What does this tell you about what the model has learned? Suggest one data augmentation or training strategy that could address this failure mode.

Ans–
This reveals that: A correctly classified cat model focuses on cat's face (correct). A misclassified cat (actuall dog) model focuses on background. This tells us the model has learned a spurious correlation (shortcut learning) like "cats appear on certain backgrounds, dogs on others".

To fix this: 1. Random Erasing to force the model to learn the foreground object. 2. Collect more diverse training examples of cats in different backgrounds.

Q2. Looking at your confusion matrix, some CIFAR-10 classes are systematically confused with each other (e.g., 'cat' - 'dog', 'automobile' - 'truck'). Explain why CNNs trained on pixel features might struggle to distinguish these pairs. What architectural change or additional input modality might help?

Ans–
These pairs share visual features at the PIXEL level. Cat & Dog both have similar fur textures, four-legged, and ears. Automobile & Truck both are vehicles with wheels, metallic surfaces, windows.

How to help: Use attention mechanisms, add input modalities (like depth info), or use larger input resolutions so finer details are preserved.

Q3. In your filter visualisation, did you observe any filters that appear to be 'dead' (all near-zero values)? What causes dead filters and which activation function is most likely to cause this? Name one remedy.

Ans–
A dead filter has weights all near zero it outputs nearly 0 regardless of input.
Cause: The dying ReLU problem where neurons consistently receive negative inputs and gradient stops flowing. Solution: Leaky ReLU, better weight initialization (He initialization), or batch normalisation before the activation function.

8 *Task 5: Transfer Learning & Fine-Tuning*

8.1 *Feature Extraction with a Frozen Base*

[19]:
```
print("Loading and preparing CIFAR-10 (48x48 upsampled) for VGG16")
(cx_tr_raw, cy_tr_raw), (cx_te_raw, cy_te_raw) = cifar10.load_data()
cx_tr_96 = tf.image.resize(cx_tr_raw, (48, 48)).numpy().astype(np.float32)
cx_te_96 = tf.image.resize(cx_te_raw, (48, 48)).numpy().astype(np.float32)

cx_tr_96 = vgg_preprocess(cx_tr_96)
cx_te_96 = vgg_preprocess(cx_te_96)

print(f"Training set: {cx_tr_96.shape}")
print(f"Test set:     {cx_te_96.shape}")

def count_params(model):
  train = sum(np.prod(v.shape) for v in model.trainable_variables)
  non_train = sum(np.prod(v.shape) for v in model.non_trainable_variables)
  return train, non_train


base_model = VGG16(weights='imagenet', include_top=False, input_shape=(48, 48,␣
 ↪3))
base_model.trainable = False

model_frozen = models.Sequential([
  base_model,
  layers.GlobalAveragePooling2D(name='gap'),
  layers.Dense(256, activation='relu', name='dense1'),
  layers.Dropout(0.5, name='drop1'),
  layers.Dense(10, activation='softmax', name='output'),
],name='frozen_vgg16')

model_frozen.compile(
  optimizer=keras.optimizers.Adam(learning_rate=0.001),
  loss='categorical_crossentropy',
  metrics=['accuracy']
)

print("\nTraining frozen model for 10 epochs")
h_frozen = model_frozen.fit(
```

```
  cx_tr_96, cifar10_train_y,
  epochs=10, batch_size=32,
  validation_split=0.1, verbose=1
)

best_frozen_acc = max(h_frozen.history['val_accuracy'])
print(f"\nBest validation accuracy: {best_frozen_acc:.4f}")
```

```
Loading and preparing CIFAR-10 (48x48 upsampled) for VGG16
Training set: (50000, 48, 48, 3)
Test set:     (10000, 48, 48, 3)
Downloading data from https://storage.googleapis.com/tensorflow/keras-
applications/vgg16/vgg16_weights_tf_dim_ordering_tf_kernels_notop.h5
58889256/58889256             1s
0us/step

Training frozen model for 10 epochs
Epoch 1/10
1407/1407             32s 20ms/step -
accuracy: 0.5070 - loss: 2.1662 - val_accuracy: 0.6446 - val_loss: 1.0356
Epoch 2/10
1407/1407             24s 17ms/step -
accuracy: 0.6114 - loss: 1.1474 - val_accuracy: 0.6742 - val_loss: 0.9585
Epoch 3/10
1407/1407             23s 17ms/step -
accuracy: 0.6421 - loss: 1.0557 - val_accuracy: 0.6810 - val_loss: 0.9452
Epoch 4/10
1407/1407             23s 17ms/step -
accuracy: 0.6562 - loss: 1.0104 - val_accuracy: 0.6724 - val_loss: 0.9514
Epoch 5/10
1407/1407             24s 17ms/step -
accuracy: 0.6693 - loss: 0.9729 - val_accuracy: 0.6892 - val_loss: 0.9283
Epoch 6/10
1407/1407             24s 17ms/step -
accuracy: 0.6768 - loss: 0.9484 - val_accuracy: 0.6912 - val_loss: 0.9333
Epoch 7/10
1407/1407             24s 17ms/step -
accuracy: 0.6860 - loss: 0.9276 - val_accuracy: 0.6888 - val_loss: 0.9291
Epoch 8/10
1407/1407             24s 17ms/step -
accuracy: 0.6978 - loss: 0.8891 - val_accuracy: 0.7022 - val_loss: 0.9445
Epoch 9/10
1407/1407             24s 17ms/step -
accuracy: 0.7000 - loss: 0.8734 - val_accuracy: 0.6968 - val_loss: 0.9422
Epoch 10/10
1407/1407             24s 17ms/step -
accuracy: 0.7089 - loss: 0.8511 - val_accuracy: 0.6952 - val_loss: 0.9498
```

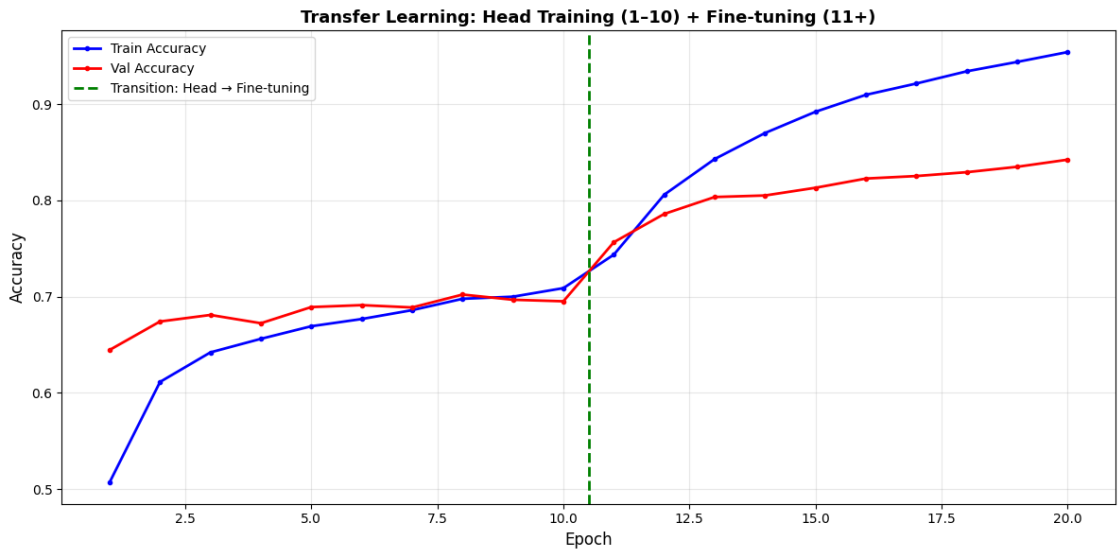
```
Best validation accuracy: 0.7022
```

8.2 *Fine-Tuning with Gradual Unfreezing*

[20]:
```
conv_layers = [l for l in base_model.layers if isinstance(l, layers.Conv2D)]
base_model.trainable = True

for layer in base_model.layers:
  if layer in conv_layers[-4:]:
      layer.trainable = True
  else:
      layer.trainable = False

model_frozen.compile(
  optimizer=keras.optimizers.Adam(learning_rate=1e-5),
  loss='categorical_crossentropy',
  metrics=['accuracy']
)

early_stop = callbacks.EarlyStopping(monitor='val_accuracy', patience=5,␣
 ↪restore_best_weights=True)

print("\nFine-tuning for 10 more epochs...")
h_finetuned = model_frozen.fit(
  cx_tr_96, cifar10_train_y,
  epochs=10, batch_size=32,
  validation_split=0.1,
  callbacks=[early_stop],
  verbose=1
)

best_ft_acc = max(h_finetuned.history['val_accuracy'])
best_ft_epoch = np.argmax(h_finetuned.history['val_accuracy']) + 11

fig,ax = plt.subplots(figsize=(12, 6))
all_val_acc = h_frozen.history['val_accuracy'] + h_finetuned.
 ↪history['val_accuracy']
all_train_acc = h_frozen.history['accuracy'] + h_finetuned.history['accuracy']
eps = range(1,len(all_val_acc) + 1)

ax.plot(eps, all_train_acc,'b-o', markersize=3,label='Train␣
 ↪Accuracy',linewidth=2)
ax.plot(eps, all_val_acc,'r-o', markersize=3,label='Val Accuracy',linewidth=2)
ax.axvline(x=10.5, color='green',linestyle='--',linewidth=2,label='Transition:␣
 ↪Head → Fine-tuning')
ax.set_xlabel('Epoch', fontsize=12)
```

```
ax.set_ylabel('Accuracy', fontsize=12)
ax.set_title('Transfer Learning: Head Training (1-10) + Fine-tuning␣
↪(11+)',fontsize=13, fontweight='bold')
ax.legend(fontsize=10)
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig('tl_finetuned.png', dpi=300)
plt.show()
```

```
Fine-tuning for 10 more epochs…
Epoch 1/10
1407/1407              67s 44ms/step -
accuracy: 0.7437 - loss: 0.7524 - val_accuracy: 0.7566 - val_loss: 0.7656
Epoch 2/10
1407/1407              56s 40ms/step -
accuracy: 0.8060 - loss: 0.5574 - val_accuracy: 0.7860 - val_loss: 0.7085
Epoch 3/10
1407/1407              56s 40ms/step -
accuracy: 0.8431 - loss: 0.4410 - val_accuracy: 0.8036 - val_loss: 0.6728
Epoch 4/10
1407/1407              56s 40ms/step -
accuracy: 0.8701 - loss: 0.3594 - val_accuracy: 0.8052 - val_loss: 0.6989
Epoch 5/10
1407/1407              56s 40ms/step -
accuracy: 0.8922 - loss: 0.2931 - val_accuracy: 0.8132 - val_loss: 0.6856
Epoch 6/10
1407/1407              56s 40ms/step -
accuracy: 0.9098 - loss: 0.2488 - val_accuracy: 0.8228 - val_loss: 0.6780
Epoch 7/10
1407/1407              56s 40ms/step -
accuracy: 0.9216 - loss: 0.2104 - val_accuracy: 0.8254 - val_loss: 0.6777
Epoch 8/10
1407/1407              56s 40ms/step -
accuracy: 0.9343 - loss: 0.1796 - val_accuracy: 0.8294 - val_loss: 0.7179
Epoch 9/10
1407/1407              56s 40ms/step -
accuracy: 0.9440 - loss: 0.1533 - val_accuracy: 0.8350 - val_loss: 0.7033
Epoch 10/10
1407/1407              56s 40ms/step -
accuracy: 0.9542 - loss: 0.1266 - val_accuracy: 0.8424 - val_loss: 0.7061
```



8.2.1 *Unfreezing Ablation Study*

[21]:
```
ablation_configs = {'Top 2 layers': 2, 'Top 8 layers': 8, 'All layers':␣
↪len(conv_layers)}
ablation_results = {}

for config_name, n_unfreeze in ablation_configs.items():
  print(f"\nConfig: {config_name} ({n_unfreeze} conv layers unfrozen)")
  base = VGG16(weights='imagenet', include_top=False, input_shape=(48, 48, 3))
  base.trainable = True
  conv_layers_local = [l for l in base.layers if isinstance(l, layers.Conv2D)]
  for layer in base.layers:
      layer.trainable = (layer in conv_layers_local[-n_unfreeze:])

  ablation_model = models.Sequential([
      base,
      layers.GlobalAveragePooling2D(),
      layers.Dense(256, activation='relu'),
      layers.Dropout(0.5),
      layers.Dense(10, activation='softmax'),
    ])

  ablation_model.compile(optimizer=keras.optimizers.
↪Adam(1e-5),loss='categorical_crossentropy',metrics=['accuracy'])
  tr_params, frozen_params = count_params(ablation_model)
```

```
  h = ablation_model.
 ↪fit(cx_tr_96,cifar10_train_y,epochs=5,batch_size=32,validation_split=0.
 ↪1,verbose=0)

  train_acc, val_acc = h.history['accuracy'][-1], max(h.history['val_accuracy'])
  gap = train_acc - val_acc
  ablation_results[config_name] = {'trainable': tr_params, 'val_acc': val_acc,␣
 ↪'gap': gap, 'overfit': "YES" if gap > 0.05 else "NO"}
  print(f"Val acc: {val_acc:.4f} | Gap: {gap:.4f} | Overfit:␣
 ↪{ablation_results[config_name]['overfit']}")

print("\nblation Study Results->")
print(f"{'Config':<20} | {'Trainable Params':>16} | {'Val Acc':>8} | {'Overfit?
 ↪':>8}")
for name, res in ablation_results.items():
  print(f"{name:<20} | {res['trainable']:>16,} | {res['val_acc']:>8.4f} |␣
 ↪{res['overfit']:>8}")
```

```
Config: Top 2 layers (2 conv layers unfrozen)
Val acc: 0.7082 | Gap: -0.0332 | Overfit: NO

Config: Top 8 layers (8 conv layers unfrozen)
Val acc: 0.8654 | Gap: 0.0258 | Overfit: NO

Config: All layers (13 conv layers unfrozen)
Val acc: 0.8632 | Gap: 0.0261 | Overfit: NO

blation Study Results->
Config               | Trainable Params |  Val Acc | Overfit?
Top 2 layers         |        4,853,514 |   0.7082 |       NO
Top 8 layers         |       14,293,258 |   0.8654 |       NO
All layers           |       14,848,586 |   0.8632 |       NO
```

8.3 *Final Benchmark — Scratch vs Transfer Learning*

[22]:
```
scratch_test_loss,scratch_test_acc = model.evaluate(cifar10_test_x,␣
 ↪cifar10_test_y, verbose=0)
frozen_test_loss,frozen_test_acc = model_frozen.evaluate(cx_te_96,␣
 ↪cifar10_test_y, verbose=0)

trainable_ft, _ = count_params(model_frozen)
scratch_tr_params, _ = count_params(model)

print("FINAL BENCHMARK TABLE")
print(f"{'Model':<35} | {'Test Acc':>9} | {'Trainable Params':>16}")
models_compared = [
```

```
  ("Scratch CNN (Dropout+BN)",    scratch_test_acc, scratch_tr_params),
  ("Fine-tuned VGG16 (last 4 CV)", frozen_test_acc,  trainable_ft),
]
for name, acc, params in models_compared:
  print(f"{name:<35} | {acc:>9.4f} | {params:>16,}")
```

```
FINAL BENCHMARK TABLE
Model                               |  Test Acc | Trainable Params
Scratch CNN (Dropout+BN)            |    0.7082 |          129,290
Fine-tuned VGG16 (last 4 CV)        |    0.8366 |        9,573,130
```

8.3.1 Analysis Questions

Q1. Explain the concept of 'negative transfer'. Under what conditions might using ImageNet pre-trained weights actually hurt performance rather than help? Give a concrete example of a domain where you would expect negative transfer.

Ans–
Negative transfer occurs when pre-trained features are so different from the target domain that initialising with them is worse than random initialisation. *Example:* Using natural photo weights (ImageNet) for grayscale X-ray classification. The learned texture filters (fur, feathers) can introduce noise instead of useful priors.

Q2. In your ablation (Problem 3), unfreezing all layers likely led to more overfitting than unfreezing only the top few. Explain the bias-variance trade-off at play here. Why do the lower layers of a CNN trained on ImageNet generalise better than the upper layers?

Ans– * Bias: A fully frozen model has high bias (underfitting) if iamgenet features don't perfectly map to the target domain. * Variance: A completely unfrozen model has high variance (overfitting) because updating 138 Million parameters on small datasets is unstable. * Sweet Spot: Unfreezing only the last few conv layers provides low bias and low variance (preserves base universal edge).

Q3. Your benchmark (Problem 4) compares parameter counts and accuracy across models. In a real deployment scenario — e.g., a mobile app — what other factors beyond accuracy would influence your choice of model? Name at least three and explain each.

Ans– 1. Inference Speed (Latency): Smaller models sacrifice some accuracy for real-time 10-50ms latency. 2. Model Size: Heavy 500MB models can't be easily shipped to edge devices. They need pruning. 3. Power consumption: Large models drain batteries quickly and generate massive heat.

Q4. Suppose you have a completely new medical imaging dataset (X-ray scans, grayscale, 512×512) with only 500 labelled training examples. Write a step-by-step transfer learning strategy you would follow, justifying every choice (which base model, how many layers to freeze, learning rate, augmentation, etc.).

Ans– 1. Handle channels (repeat grayscale 3 times). 2. Start by completely freezing the base layer to prevent catastrophic overfitting. 3. Train only a small classification head using aggressive data augmentation. 4. After head converges, carefully unfreeze the last 2-3 blocks using an extremely tiny learning rate (1e-5). 5. Always rely on k-fold cross-validation when data is sparse.